

FIT9137 – Computer Architecture ASSIGNMENT 3

2025 – Semester 1

Name: Bharath Arun Gandhimani

Student ID: 35501308

TASK 1- Routing

Optimal Path Selection:

R1 -> R2

There are three paths to R2 from R1.

Path 1: R1 -> R2 (Direct)

There is a direct connection between R1 and R2 through eth3.

Delay = 100us

Path 2: R1 -> R3 -> R2

R1 can reach R2 through R3.

Delay = 100us (R1 -> R3) + 110us (R3 -> R2) = 210us > Path 1.

Path 3: R1 -> R4 -> R2

R1 can reach R2 through R4.

Delay = 110us (R1 -> R4) + 110us (R4 -> R2) = 220us > Path 1.

Therefore, Optimal Path for R1 to R2 is direct via eth3.

R1 -> R3

There are three paths to R3 from R1.

Path 1: R1 -> R3 (Direct)

There is a direct connection between R1 and R3 through eth1

Delay = 100us

Path 2: R1 -> R2 -> R3

R1 can reach R3 through R2.

Delay = 100us (R1 -> R2) + 110us (R2 -> R3) = 210us > Path 1.

Path 3: R1 -> R4 -> R2 -> R3

R1 can reach R3 through R4 and from R4 to R2 and then from R2 to R3.

Delay = 110us (R1 -> R4) + 110us (R4 -> R2) + 110us (R2 -> R3) = 330us > Path 1

Therefore, Optimal Path for R1 to R3 is direct via eth1.

R1 to R4

There are three paths to R4 from R1.

Path 1: R1 -> R4 (Direct)

There is a direct connection between R1 and R4 through eth2.

Delay = 110us

Path 2: R1 -> R2 -> R4

R1 can reach R4 through R2.

Delay = 100us (R1 -> R2) + 110 us (R2 -> R4) = 210us > Path 1.

Path 3: R1 -> R3 -> R2 -> R4

R1 can reach R4 through R3 and from R3 to R2 and then from R2 to R4.

Delay = 100us (R1 -> R3) + 110us (R3 -> R2) + 110us (R2 -> R4) = 320us > Path 1.

Therefore, Optimal Path for R1 to R4 is direct via eth2.

R2 to R1

The path from R2 to R1 is the same path as R1 to R2.

Therefore, Optimal Path for R2 to R1 is direct via eth3.

R2 to R3

There are three paths to R3 from R2.

Path 1: R2 -> R3 (Direct)

There is a direct connection between R2 and R3 through eth1.

Delay = 110us.

Path 2: R2 -> R1 -> R3

R2 can reach R3 through R1.

Delay = 100us (R2 -> R1) + 100us (R1 -> R3) = 200us > Path 1

Path 3: R2 -> R4 -> R1 -> R3

R2 can reach R3 through R4 and then from R4 to R1 and then from R1 to R3.

Delay = 110us (R2 -> R4) + 110us (R4 -> R1) + 100us (R1 -> R3) = 320us > Path 1

Therefore, Optimal Path for R2 to R3 is direct via eth1.

R2 to R4

There are three paths to R4 from R2.

Path 1: R2 -> R4 (Direct)

There is a direct connection between R2 and R4 through eth2.

Delay = 110us.

Path 2: R2 -> R1 -> R4

R2 can reach R4 through R1.

Delay = 100us (R2 -> R1) + 110us (R1 -> R4) = 210us > Path 1.

Path 3: R2 -> R3 -> R1 -> R4

R2 can reach R4 through R3 and then from R3 to R1 and then from R1 to R4.

Delay = 110us (R2 -> R3) + 100us (R3 -> R1) + 110us (R1 -> R4) = 320 us > Path 1.

Therefore, Optimal Path for R2 to R4 is direct via eth2.

R3 to R1

The path from R3 to R1 is the same path as R1 to R3.

Therefore, Optimal Path for R3 to R1 is direct via eth1.

R3 to R2

The path from R3 to R2 is the same path as R2 to R3.

Therefore, Optimal Path for R3 to R2 is direct via eth2.

R3 to R4

There are two paths to R4 from R3.

Path 1: R3 -> R1 -> R4

R3 can reach R4 through R1.

Delay = 100us (R3 -> R1) + 110us (R1 -> R4) = 210us.

Path 2: R3 -> R2 -> R4

R3 can reach R4 through R2.

Delay = 110us (R3 -> R2) + 110us (R2 -> R4) = 220 us > Path 1.

Therefore, Optimal Path for R3 to R4 is via R1.
--

R4 to R1

The path from R4 to R1 is the same path as R1 to R4.

Therefore, Optimal Path for R4 to R1 is direct via eth2.

R4 to R2

The path from R4 to R2 is the same path as R2 to R4.

Therefore, Optimal Path for R4 to R2 is direct via eth1.

R4 to R3

There are two paths to R3 from R4.

Path 1: R4 -> R1 -> R3

R4 can reach R3 through R1.

Delay = 110us (R4 -> R1) + 100us (R1 -> R3) = 210us.

Path 2: R4 -> R2 -> R3

R4 can reach R3 through R2.

Delay = 110us (R4 -> R2) + 110us (R2 -> R3) = 220us > Path 1.

Therefore, Optimal Path for R4 to R3 is via R1.
--

Internet Router

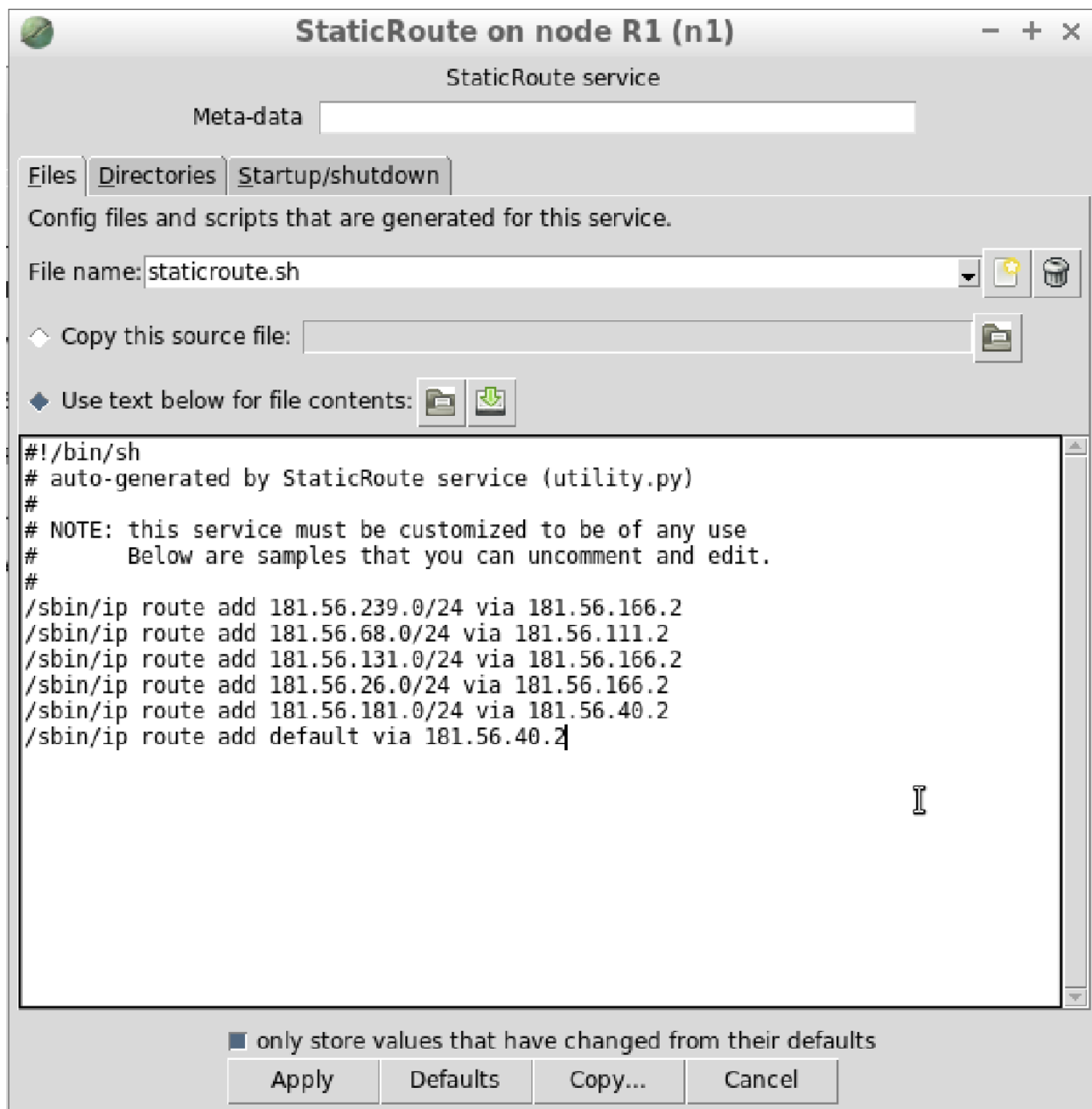
This router acts as the central hub. It needs to know how to reach the entire Talos network and the entire Delos network.

Minerva Router

Minerva needs to reach the Talos network and the global DNS server. It's default gateway is the Internet router for all external traffic.

Static Route Configuration

Node R1:



The image shows a window titled "StaticRoute on node R1 (n1)" with a subtitle "StaticRoute service". It has a "Meta-data" field at the top. Below it are three tabs: "Files", "Directories", and "Startup/shutdown", with "Files" selected. The main area is labeled "Config files and scripts that are generated for this service." and contains a "File name:" field with "staticroute.sh". There are icons for "Copy this source file:" and "Use text below for file contents:". The "Use text below for file contents:" option is selected, and a large text area contains the following script:

```
#!/bin/sh
# auto-generated by StaticRoute service (utility.py)
#
# NOTE: this service must be customized to be of any use
#       Below are samples that you can uncomment and edit.
#
/sbin/ip route add 181.56.239.0/24 via 181.56.166.2
/sbin/ip route add 181.56.68.0/24 via 181.56.111.2
/sbin/ip route add 181.56.131.0/24 via 181.56.166.2
/sbin/ip route add 181.56.26.0/24 via 181.56.166.2
/sbin/ip route add 181.56.181.0/24 via 181.56.40.2
/sbin/ip route add default via 181.56.40.2
```

At the bottom, there is a checkbox labeled "only store values that have changed from their defaults" which is checked. Below the checkbox are four buttons: "Apply", "Defaults", "Copy...", and "Cancel".

Node R2:

 **StaticRoute on node R2 (n2)** - + x

StaticRoute service

Meta-data

Files Directories Startup/shutdown

Config files and scripts that are generated for this service.

File name:  

◆ Copy this source file: 

◆ Use text below for file contents:  

```
#!/bin/sh
# auto-generated by StaticRoute service (utility.py)
#
# NOTE: this service must be customized to be of any use
#       Below are samples that you can uncomment and edit.
#
/sbin/ip route add 181.56.18.0/24 via 181.56.166.1
/sbin/ip route add 181.56.40.0/24 via 181.56.26.1
/sbin/ip route add 181.56.11.0/24 via 181.56.166.1
/sbin/ip route add 181.56.181.0/24 via 181.56.26.1
/sbin/ip route add 181.56.68.0/24 via 181.56.131.1
/sbin/ip route add default add 181.56.26.1
```

☒ only store values that have changed from their defaults

Node R3:

 **StaticRoute on node R3 (n3)** - + x

StaticRoute service

Meta-data

Files Directories Startup/shutdown

Config files and scripts that are generated for this service.

File name:  

◆ Copy this source file: 

◆ Use text below for file contents:  

```
#!/bin/sh
# auto-generated by StaticRoute service (utility.py)
#
# NOTE: this service must be customized to be of any use
#       Below are samples that you can uncomment and edit.
#
/sbin/ip route add 181.56.18.0/24 via 181.56.40.1
/sbin/ip route add 181.56.239.0/24 via 181.56.26.2
/sbin/ip route add 181.56.68.0/24 via 181.56.40.1
/sbin/ip route add 181.56.111.0/24 via 181.56.40.1
/sbin/ip route add 181.56.131.0/24 via 181.56.26.2
/sbin/ip route add 181.56.166./24 via 181.56.40.1
/sbin/ip route add default via 98.141.71.2
```

☒ only store values that have changed from their defaults

Node R4:

 **StaticRoute on node R4 (n4)** - + x

StaticRoute service

Meta-data

Files Directories Startup/shutdown

Config files and scripts that are generated for this service.

File name:  

◆ Copy this source file: 

◆ Use text below for file contents:  

```
#!/bin/sh
# auto-generated by StaticRoute service (utility.py)
#
# NOTE: this service must be customized to be of any use
#       Below are samples that you can uncomment and edit.
#
/sbin/ip route add 181.56.18.0/24 via 181.56.111.1
/sbin/ip route add 181.56.40.0/24 via 181.56.111.1
/sbin/ip route add 181.56.239.0/24 via 181.56.131.2
/sbin/ip route add 181.56.181.0/24 via 181.56.111.1
/sbin/ip route add 181.56.166.0/24 via 181.56.111.1
/sbin/ip route add 181.56.26.0/24 via 181.56.131.2
/sbin/ip route add default via 181.56.131.2
```

☒ only store values that have changed from their defaults

Internet Router:

 **StaticRoute on node Internet (n8)** - + x

StaticRoute service

Meta-data

Files

Directories

Startup/shutdown

Config files and scripts that are generated for this service.

File name:  

☐ Copy this source file: 

☒ Use text below for file contents:  

```
#!/bin/sh
# auto-generated by StaticRoute service (utility.py)
#
# NOTE: this service must be customized to be of any use
#       Below are samples that you can uncomment and edit.
#
/sbin/ip route add 181.56.0.0/16 via 98.141.71.1
/sbin/ip route add 114.181.0.0/16 via 48.198.245.2
```

☒ only store values that have changed from their defaults

Apply

Defaults

Copy...

Cancel

Minerva:



StaticRoute on node minerva (n18)

- + x

StaticRoute service

Meta-data

Files

Directories

Startup/shutdown

Config files and scripts that are generated for this service.

File name:  

◆ Copy this source file: 

◆ Use text below for file contents:  

```
#!/bin/sh
# auto-generated by StaticRoute service (utility.py)
#
# NOTE: this service must be customized to be of any use
#       Below are samples that you can uncomment and edit.
#
/sbin/ip route add default via 48.198.245.1
```

☒ only store values that have changed from their defaults

Apply

Defaults

Copy...

Cancel

TASK 1 - Testing

Test 1: Talos Internal → Internal

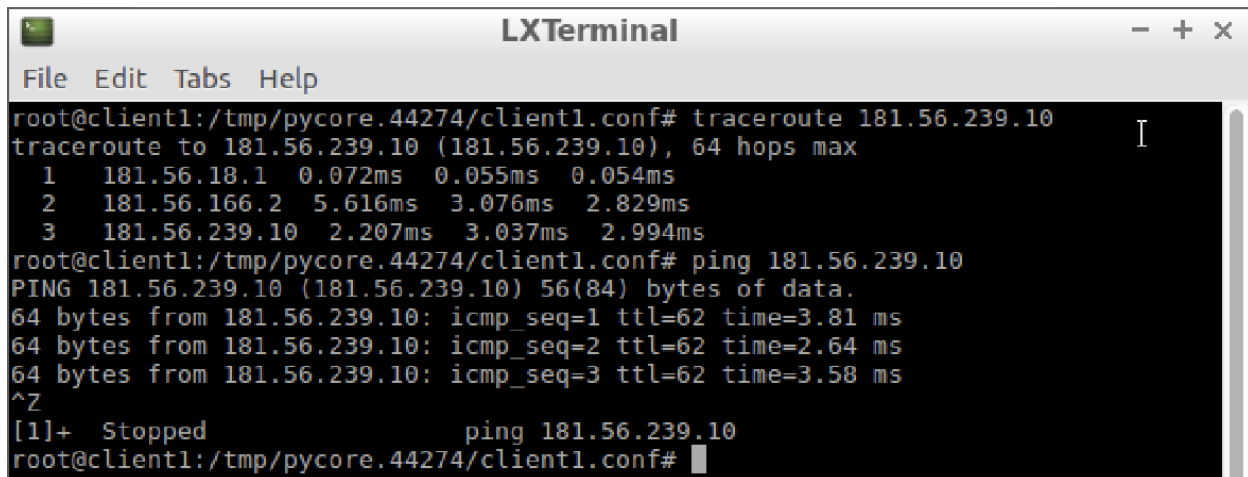
From client1 (via R1) to intranet (via R2)

Command:

ping 181.56.239.10

tracert 181.56.239.10

Expected path: client1 → R1 (181.56.18.1) → R2 (181.56.166.2) → intranet (181.56.239.10)

A screenshot of an LXTerminal window. The title bar says "LXTerminal" with standard window controls. The menu bar shows "File Edit Tabs Help". The terminal content shows a user at the prompt "root@client1:/tmp/pycore.44274/client1.conf#" running "tracert 181.56.239.10". The output shows three hops: 1 (181.56.18.1), 2 (181.56.166.2), and 3 (181.56.239.10) with associated RTT values. Then the user runs "ping 181.56.239.10", showing three successful ping responses with 56(84) bytes of data and various times. The session ends with a Ctrl-Z (^Z) and a message "[1]+ Stopped ping 181.56.239.10".

```
root@client1:/tmp/pycore.44274/client1.conf# tracert 181.56.239.10
tracert to 181.56.239.10 (181.56.239.10), 64 hops max
 1  181.56.18.1  0.072ms  0.055ms  0.054ms
 2  181.56.166.2  5.616ms  3.076ms  2.829ms
 3  181.56.239.10  2.207ms  3.037ms  2.994ms
root@client1:/tmp/pycore.44274/client1.conf# ping 181.56.239.10
PING 181.56.239.10 (181.56.239.10) 56(84) bytes of data.
64 bytes from 181.56.239.10: icmp_seq=1 ttl=62 time=3.81 ms
64 bytes from 181.56.239.10: icmp_seq=2 ttl=62 time=2.64 ms
64 bytes from 181.56.239.10: icmp_seq=3 ttl=62 time=3.58 ms
^Z
[1]+  Stopped                  ping 181.56.239.10
root@client1:/tmp/pycore.44274/client1.conf#
```

Shows R1 → R2 path is functional.

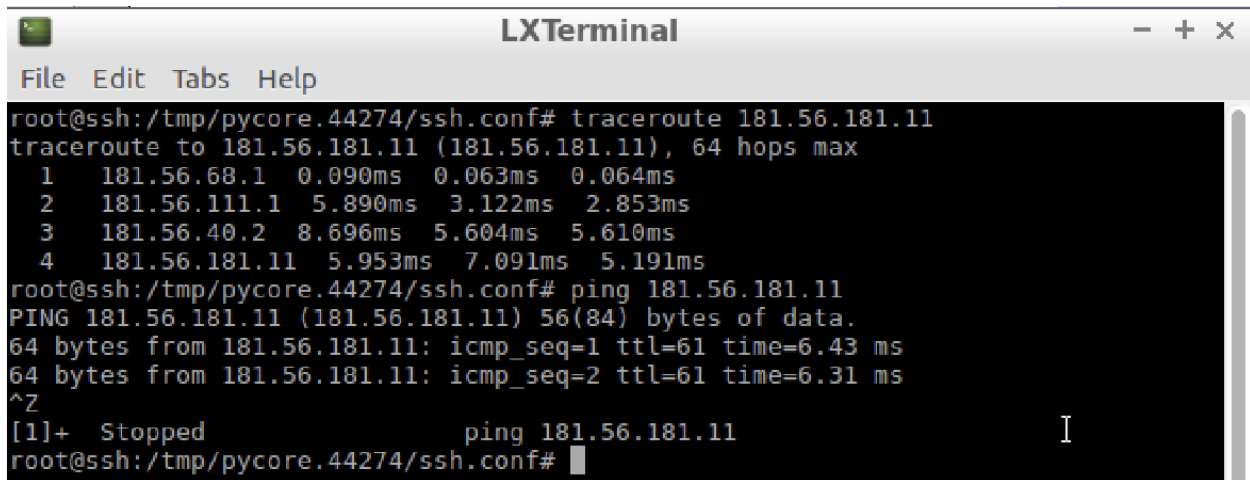
From ssh (via R4) to web server (via R3)

Command:

ping 181.56.181.11

traceroute 181.56.181.11

Expected path: ssh → R4 (181.56.68.1) → R1 (181.56.111.1) → R3 (181.56.40.2) → web

A screenshot of an LXTerminal window. The title bar says "LXTerminal" with standard window controls. The menu bar shows "File", "Edit", "Tabs", and "Help". The terminal content shows a user at a root prompt on an SSH connection. They run "traceroute 181.56.181.11", which shows a path of 4 hops: 181.56.68.1, 181.56.111.1, 181.56.40.2, and 181.56.181.11. Then they run "ping 181.56.181.11", which shows two successful pings with times around 6.4ms and 6.3ms. The prompt then shows "[1]+ Stopped ping 181.56.181.11" and the user returns to the root prompt.

```
root@ssh:/tmp/pycore.44274/ssh.conf# traceroute 181.56.181.11
traceroute to 181.56.181.11 (181.56.181.11), 64 hops max
 1  181.56.68.1  0.090ms  0.063ms  0.064ms
 2  181.56.111.1  5.890ms  3.122ms  2.853ms
 3  181.56.40.2  8.696ms  5.604ms  5.610ms
 4  181.56.181.11  5.953ms  7.091ms  5.191ms
root@ssh:/tmp/pycore.44274/ssh.conf# ping 181.56.181.11
PING 181.56.181.11 (181.56.181.11) 56(84) bytes of data.
64 bytes from 181.56.181.11: icmp_seq=1 ttl=61 time=6.43 ms
64 bytes from 181.56.181.11: icmp_seq=2 ttl=61 time=6.31 ms
^Z
[1]+  Stopped                  ping 181.56.181.11
root@ssh:/tmp/pycore.44274/ssh.conf#
```

Confirms **R4 → R3 routing via R1** is in place.

Test 2: Internal → DMZ (R3)

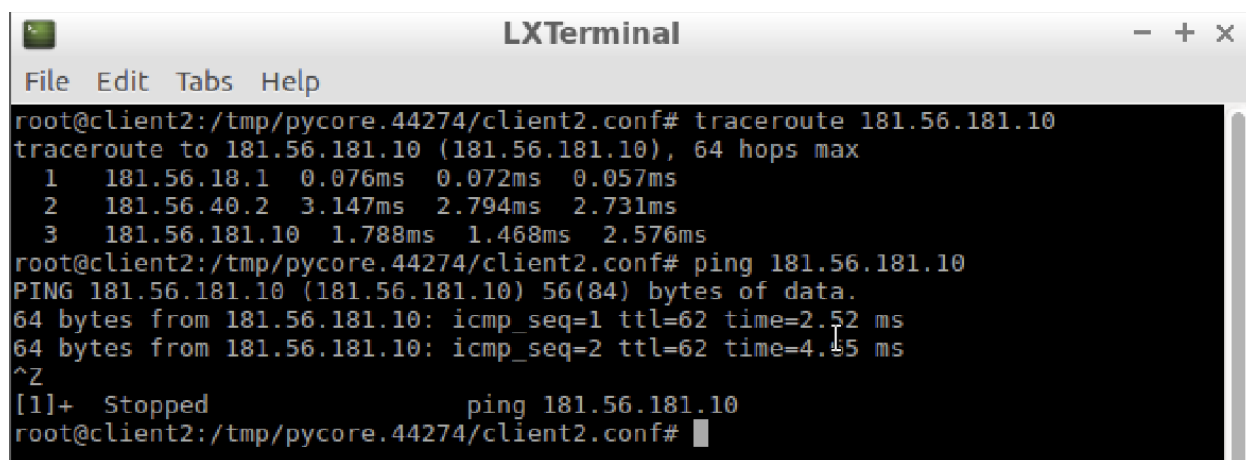
From client2 (R1 client) to dns (DMZ)

Commands:

ping 181.56.181.10

tracert 181.56.181.10

Expected path: client2 → R1 → R3 → dns (181.56.181.10)

A screenshot of an LXTerminal window. The title bar says "LXTerminal" and has standard window controls. The menu bar shows "File Edit Tabs Help". The terminal content shows a user at the prompt "root@client2:/tmp/pycore.44274/client2.conf#" running "tracert 181.56.181.10". The output shows three hops: 1 (181.56.18.1), 2 (181.56.40.2), and 3 (181.56.181.10). After the traceroute, the user runs "ping 181.56.181.10". The output shows two successful ping requests with 64 bytes of data, showing TTL=62 and response times of 2.52 ms and 4.55 ms. The terminal ends with a carriage return and the prompt again.

```
root@client2:/tmp/pycore.44274/client2.conf# tracert 181.56.181.10
tracert to 181.56.181.10 (181.56.181.10), 64 hops max
 1  181.56.18.1  0.076ms  0.072ms  0.057ms
 2  181.56.40.2  3.147ms  2.794ms  2.731ms
 3  181.56.181.10 1.788ms  1.468ms  2.576ms
root@client2:/tmp/pycore.44274/client2.conf# ping 181.56.181.10
PING 181.56.181.10 (181.56.181.10) 56(84) bytes of data.
64 bytes from 181.56.181.10: icmp_seq=1 ttl=62 time=2.52 ms
64 bytes from 181.56.181.10: icmp_seq=2 ttl=62 time=4.55 ms
^Z
[1]+  Stopped                  ping 181.56.181.10
root@client2:/tmp/pycore.44274/client2.conf#
```

Confirms Talos → DMZ reachability and routing through R3.

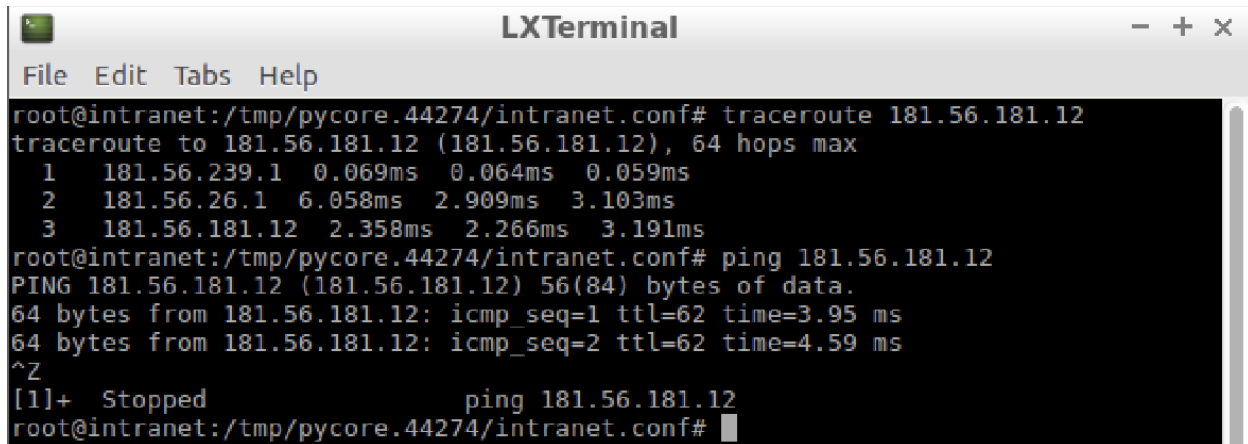
From intranet to mail server

Commands:

ping 181.56.181.12

traceroute 181.56.181.12

Expected path: intranet → R2 → R3 → mail

A screenshot of an LXTerminal window. The title bar says "LXTerminal" and has standard window controls. The menu bar shows "File Edit Tabs Help". The terminal content shows a user at the prompt "root@intranet:/tmp/pycore.44274/intranet.conf#" running "traceroute 181.56.181.12". The output shows three hops: 1 (181.56.239.1), 2 (181.56.26.1), and 3 (181.56.181.12). Then the user runs "ping 181.56.181.12", showing two successful pings with times of 3.95 ms and 4.59 ms. The session ends with a Ctrl-Z (^Z) and a "[1]+ Stopped ping 181.56.181.12" message.

```
root@intranet:/tmp/pycore.44274/intranet.conf# traceroute 181.56.181.12
traceroute to 181.56.181.12 (181.56.181.12), 64 hops max
 1  181.56.239.1  0.069ms  0.064ms  0.059ms
 2  181.56.26.1  6.058ms  2.909ms  3.103ms
 3  181.56.181.12  2.358ms  2.266ms  3.191ms
root@intranet:/tmp/pycore.44274/intranet.conf# ping 181.56.181.12
PING 181.56.181.12 (181.56.181.12) 56(84) bytes of data.
64 bytes from 181.56.181.12: icmp_seq=1 ttl=62 time=3.95 ms
64 bytes from 181.56.181.12: icmp_seq=2 ttl=62 time=4.59 ms
^Z
[1]+  Stopped                  ping 181.56.181.12
root@intranet:/tmp/pycore.44274/intranet.conf#
```

Test 3: DMZ → Internal (R3 back to others)

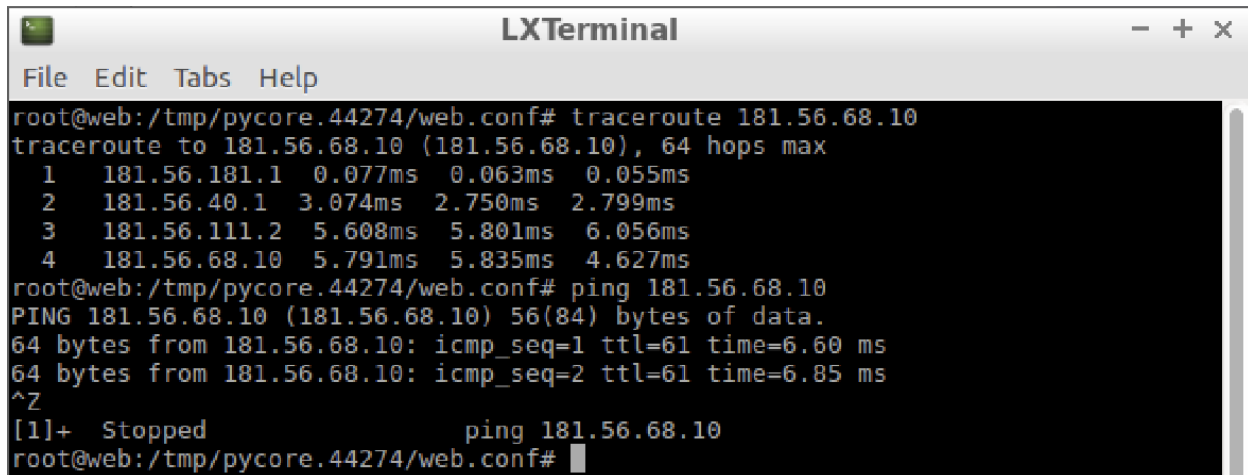
From web (181.56.181.11) to ssh server (181.56.68.10)

Commands:

ping 181.56.68.10

traceroute 181.56.68.10

Expected path: web → R3 → R1 → R4 → ssh

A screenshot of an LXTerminal window. The title bar says "LXTerminal" with standard window controls. The menu bar has "File", "Edit", "Tabs", and "Help". The terminal content shows a user at a prompt "root@web:/tmp/pycore.44274/web.conf#" running "traceroute 181.56.68.10". The output shows a path of 4 hops: 1 (181.56.181.1), 2 (181.56.40.1), 3 (181.56.111.2), and 4 (181.56.68.10). After the traceroute, the user runs "ping 181.56.68.10". The output shows two successful ping requests with 56(84) bytes of data, each taking approximately 6.6-6.85 ms. The terminal ends with a prompt and a cursor.

```
root@web:/tmp/pycore.44274/web.conf# traceroute 181.56.68.10
traceroute to 181.56.68.10 (181.56.68.10), 64 hops max
 1  181.56.181.1  0.077ms  0.063ms  0.055ms
 2  181.56.40.1  3.074ms  2.750ms  2.799ms
 3  181.56.111.2  5.608ms  5.801ms  6.056ms
 4  181.56.68.10  5.791ms  5.835ms  4.627ms
root@web:/tmp/pycore.44274/web.conf# ping 181.56.68.10
PING 181.56.68.10 (181.56.68.10) 56(84) bytes of data.
64 bytes from 181.56.68.10: icmp_seq=1 ttl=61 time=6.60 ms
64 bytes from 181.56.68.10: icmp_seq=2 ttl=61 time=6.85 ms
^Z
[1]+  Stopped                  ping 181.56.68.10
root@web:/tmp/pycore.44274/web.conf#
```

Confirms **R3 → R4** uses **R1** path.

Test 4: Default Gateway Check (Talos uses R3)

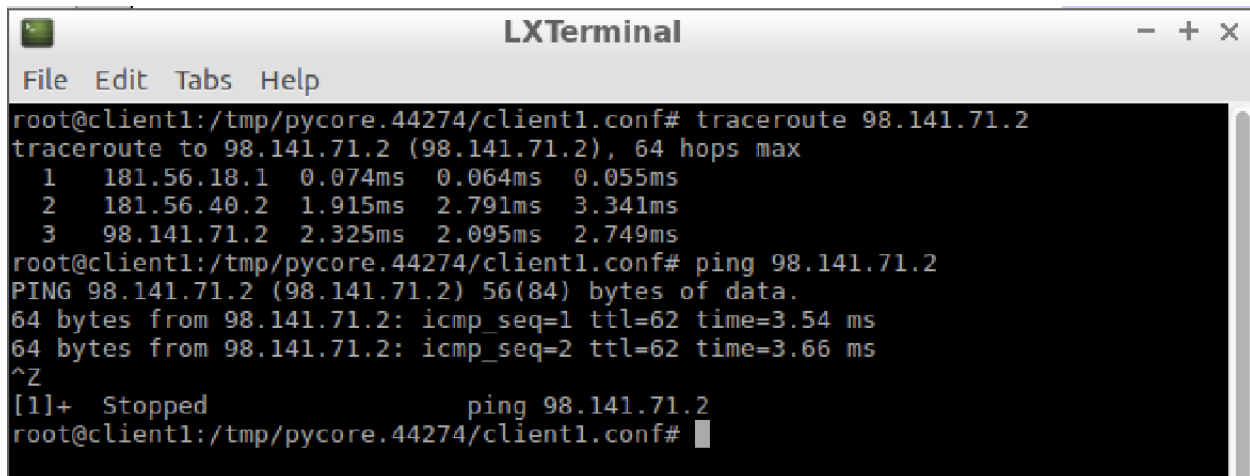
From client1 to Internet router

Commands:

ping 98.141.71.2

traceroute 98.141.71.2

Expected path: client1 → R1 → R3 → Internet (98.141.71.2)



```
LXTerminal
File Edit Tabs Help
root@client1:/tmp/pycore.44274/client1.conf# traceroute 98.141.71.2
traceroute to 98.141.71.2 (98.141.71.2), 64 hops max
 1  181.56.18.1  0.074ms  0.064ms  0.055ms
 2  181.56.40.2  1.915ms  2.791ms  3.341ms
 3  98.141.71.2  2.325ms  2.095ms  2.749ms
root@client1:/tmp/pycore.44274/client1.conf# ping 98.141.71.2
PING 98.141.71.2 (98.141.71.2) 56(84) bytes of data:
64 bytes from 98.141.71.2: icmp_seq=1 ttl=62 time=3.54 ms
64 bytes from 98.141.71.2: icmp_seq=2 ttl=62 time=3.66 ms
^Z
[1]+  Stopped                  ping 98.141.71.2
root@client1:/tmp/pycore.44274/client1.conf#
```

Confirms default route from Talos is via R3.

Test 5: R3 Default → Internet

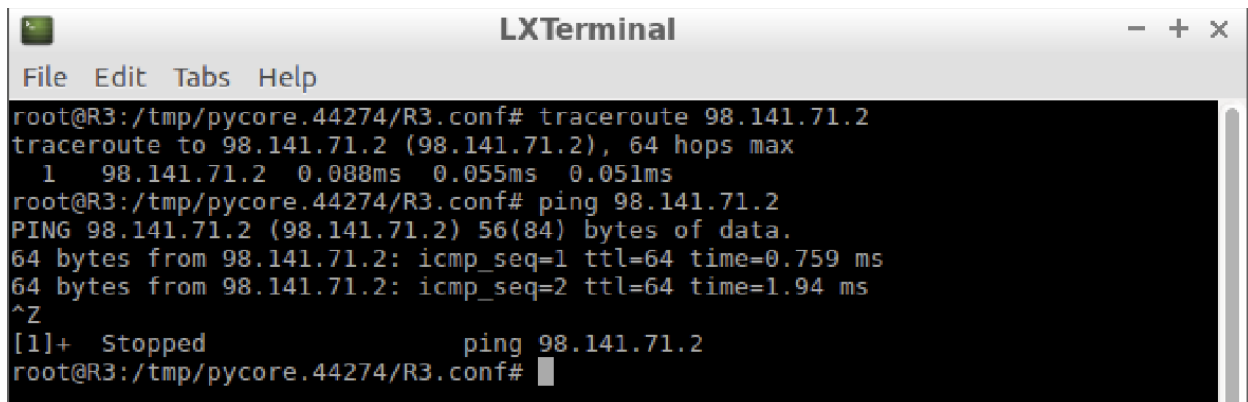
From R3 to Internet router

Commands:

ping 98.141.71.2

traceroute 98.141.71.2

Expected path: R3 (eth3: 98.141.71.1) → Internet (98.141.71.2)

A screenshot of an LXTerminal window. The title bar says "LXTerminal" with standard window controls. The menu bar has "File", "Edit", "Tabs", and "Help". The terminal text shows a user at the root of R3 running a traceroute and a ping command to 98.141.71.2. The traceroute shows a single hop with three timing measurements. The ping shows two successful requests with 56(84) bytes of data, TTL=64, and response times of 0.759 ms and 1.94 ms. The user then presses Ctrl-Z (^Z) and the terminal shows "[1]+ Stopped ping 98.141.71.2" before returning to the root prompt.

```
root@R3:/tmp/pycore.44274/R3.conf# traceroute 98.141.71.2
traceroute to 98.141.71.2 (98.141.71.2), 64 hops max
 1  98.141.71.2  0.088ms  0.055ms  0.051ms
root@R3:/tmp/pycore.44274/R3.conf# ping 98.141.71.2
PING 98.141.71.2 (98.141.71.2) 56(84) bytes of data.
64 bytes from 98.141.71.2: icmp_seq=1 ttl=64 time=0.759 ms
64 bytes from 98.141.71.2: icmp_seq=2 ttl=64 time=1.94 ms
^Z
[1]+  Stopped                  ping 98.141.71.2
root@R3:/tmp/pycore.44274/R3.conf#
```

Confirms R3's default route is working.

Test 6: Minerva Default → Internet

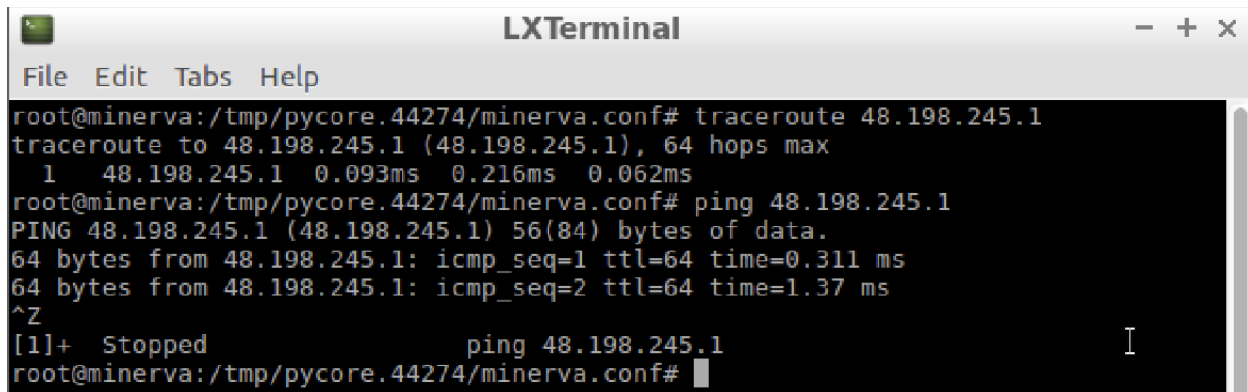
From minerva to Internet router

Commands:

ping 48.198.245.1

traceroute 48.198.245.1

Expected path: minerva (48.198.245.2) → Internet (48.198.245.1)

A screenshot of an LXTerminal window. The title bar says "LXTerminal" and has standard window controls. The menu bar includes "File", "Edit", "Tabs", and "Help". The terminal content shows a user at the "root@minerva:/tmp/pycore.44274/minerva.conf#" prompt. They run "traceroute 48.198.245.1", which shows a single hop to 48.198.245.1 with three time measurements. Then they run "ping 48.198.245.1", which shows two successful pings with 56(84) bytes of data, TTL=64, and response times of 0.311 ms and 1.37 ms. The user presses Ctrl-Z (^Z), and the terminal shows "[1]+ Stopped ping 48.198.245.1". The prompt returns to "root@minerva:/tmp/pycore.44274/minerva.conf#".

```
root@minerva:/tmp/pycore.44274/minerva.conf# traceroute 48.198.245.1
traceroute to 48.198.245.1 (48.198.245.1), 64 hops max
 1  48.198.245.1  0.093ms  0.216ms  0.062ms
root@minerva:/tmp/pycore.44274/minerva.conf# ping 48.198.245.1
PING 48.198.245.1 (48.198.245.1) 56(84) bytes of data.
64 bytes from 48.198.245.1: icmp_seq=1 ttl=64 time=0.311 ms
64 bytes from 48.198.245.1: icmp_seq=2 ttl=64 time=1.37 ms
^Z
[1]+  Stopped                  ping 48.198.245.1
root@minerva:/tmp/pycore.44274/minerva.conf#
```

Verifies default route of Minerva is correct.

TASK 2 – DHCP Server

The Delos network is indeed divided into two subnets:

1st Subnet: 114.181.127.0/24 contains apollo, artemis, and demeter.

2nd Subnet: 114.181.172.0/24 contains extClient1, extClient2, and leto. This is the subnet we need to configure DHCP.

Reason for choosing 2nd Subnet:

The DHCP server on minerva is only responsible for assigning IP addresses to clients which is directly connected subnet where dynamic allocation is required. In this scenario, that is specifically the client subnet 114.181.172.0/24.

Identifying DNS for Delos:

artemis is explicitly identified as the DNS server for the delos.edu domain.

Local DNS Resolution for Delos Network:

- When a client within the Delos network (like extClient1 or extClient2) wants to resolve a hostname like leto.delos.edu or apollo.delos.edu, it's most efficient to have a local DNS server handle these requests directly. artemis knows about all the hosts within delos.edu.

Efficiency and Reduced Traffic:

- If we were to point clients directly to clio (the global DNS server at 196.37.86.10), every single DNS request, even for hosts within delos.edu or talos.edu, would have to traverse the entire network: extClient -> minerva -> Internet -> clio.
- By pointing clients to artemis, local delos.edu queries are often resolved by artemis immediately, reducing latency and traffic on the "Internet" link.

artemis Acts as a Forwarder:

- The problem description states: "The server only resolves the names for the two domains in the configuration (talos.edu and delos.edu) by sending the request to the corresponding nameserver for each domain and sending back the response to the requesting client."
- This implies that artemis (the Delos DNS server) is configured to:
- Resolve names for delos.edu directly.
- Forward requests for talos.edu to dns.talos.edu (the Talos DNS server).
- Forward all other requests (e.g., for external domains, if they existed) to clio (the global DNS server).
- So, artemis serves as an intermediary. It's the first point of contact for Delos clients, and it knows where to go next for different types of queries.

Security and Centralization:

- Having a designated internal DNS server (artemis) allows for better control, logging, and potentially filtering of DNS requests within the Delos network. It centralizes DNS management for that organization.

Minera DHCP

DHCP on node minerva (n18)

DHCP service

Meta-data

Files | Directories | Startup/shutdown

Config files and scripts that are generated for this service.

File name:  

◆ Copy this source file: 

◆ Use text below for file contents:  

```
# Place DHCP Server configuration here

log-facility local6;

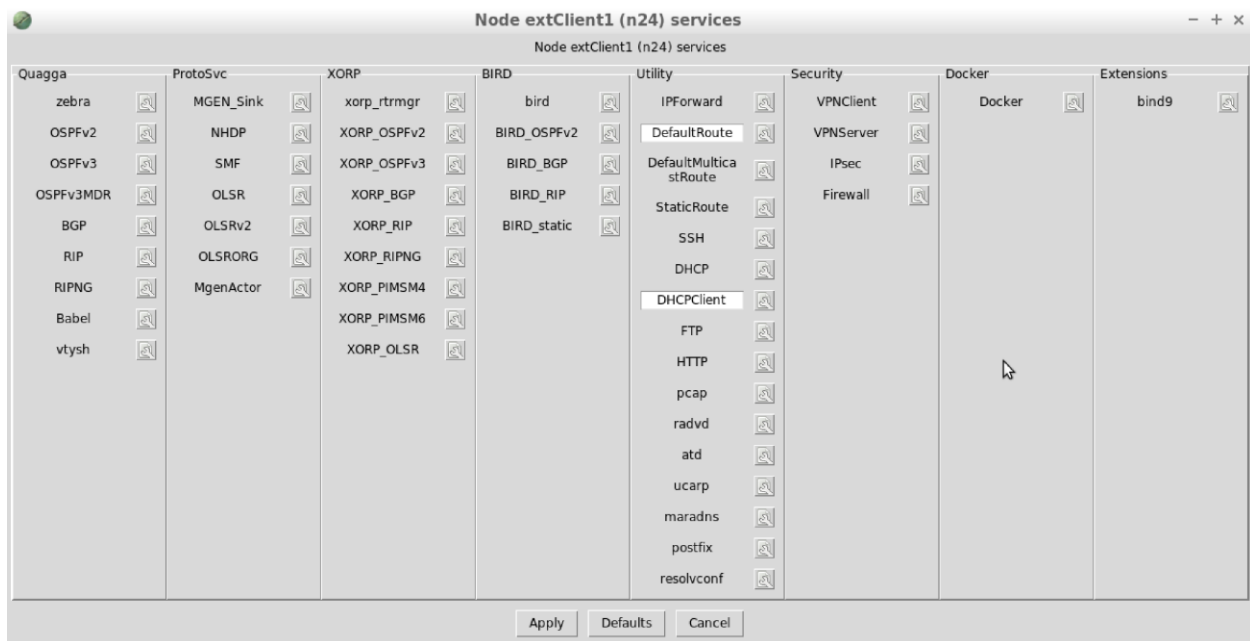
default-lease-time 36000;
max-lease-time 72000;

ddns-update-style none;

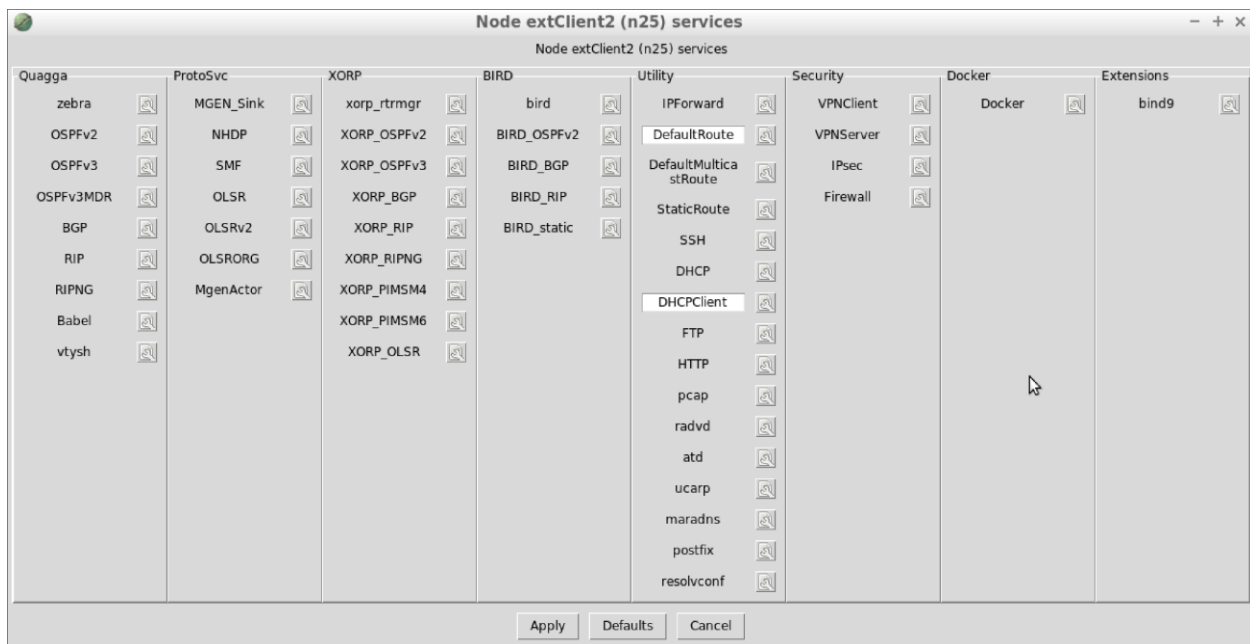
subnet 114.181.172.0 netmask 255.255.255.0 {
    pool {
        range 114.181.172.100 114.181.172.200;
        default-lease-time 36000;
        option routers 114.181.172.1;
        option domain-name-servers 114.181.127.10;
        option domain-name "delos.edu";
    }
}
```

☒ only store values that have changed from their defaults

DHCPClient enabled on extClient1



DHCPClient enabled on extClient2



TASK 2 - Testing:

1. Verify IP Address on Clients:

- **Source Node:** extClient1
- **Command:** ip a or ifconfig
- **Expected :** You should see an IP address assigned from the range you defined (e.g., 114.181.172.100-200).

```
root@extClient1:/tmp/pycore.54934/extClient1.conf# ip a show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
249: eth0@if250: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:00:00:aa:00:1d brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 114.181.172.20/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet 114.181.172.100/24 brd 114.181.172.255 scope global secondary eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:feaa:1d/64 scope link
        valid_lft forever preferred_lft forever
root@extClient1:/tmp/pycore.54934/extClient1.conf#
```

- **IP Address:** 114.181.172.100 is the first IP address in your DHCP pool range (114.181.172.100 to 114.181.172.200) as configured on minerva.
- **Netmask:** /24 is correct.
- **Secondary:** This indicates that the interface has multiple IP addresses assigned to it.
- Verify the netmask, broadcast address, and that the default gateway is set correctly.
- Repeat for extClient2.

```
LXTerminal
File Edit Tabs Help
root@extClient2:/tmp/pycore.54934/extClient2.conf# ip a show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
251: eth0@if252: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:00:00:aa:00:1e brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 114.181.172.21/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet 114.181.172.101/24 brd 114.181.172.255 scope global secondary eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:feaa:1e/64 scope link
        valid_lft forever preferred_lft forever
root@extClient2:/tmp/pycore.54934/extClient2.conf#
```

2. Verify DNS and Default Gateway:

- **Source Node:** extClient1
- **Command:** run: ip route
- **Expected:** Confirm the default route points to 114.181.172.1.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54934/extClient1.conf# ip route
default via 114.181.172.1 dev eth0
114.181.172.0/24 dev eth0 proto kernel scope link src 114.181.172.20
root@extClient1:/tmp/pycore.54934/extClient1.conf#
```

- **Source Node:** extClient1
- **Command:** run: cat /etc/resolv.conf
- **Expected:** Confirm that the nameserver entry points to 114.181.127.10 (artemis).

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54934/extClient1.conf# cat /etc/resolv.conf
# Dynamic resolv.conf(5) file for glibc resolver(3) generated by resolvconf(8)
#     DO NOT EDIT THIS FILE BY HAND -- YOUR CHANGES WILL BE OVERWRITTEN
nameserver 114.181.127.10
search delos.edu
root@extClient1:/tmp/pycore.54934/extClient1.conf#
```

3. Verify leto's Static IP:

- **Source Node:** leto
- **Command:** ip a or ifconfig.
- **Expected:** Confirm that its IP address is still the static one assigned (e.g., 114.181.172.10) and has not been changed by DHCP.

```
LXTerminal
File Edit Tabs Help
root@leto:/tmp/pycore.54934/leto.conf# ip a show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
261: eth0@if262: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:00:00:aa:00:23 brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 114.181.172.10/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:feaa:23/64 scope link
        valid_lft forever preferred_lft forever
root@leto:/tmp/pycore.54934/leto.conf#
```

4. Verify DHCP Lease Acquisition on extClient1 and extClient2

- **Objective:** Confirm that extClient1 and extClient2 successfully obtain an IP address, netmask, broadcast address, and lease information from the minerva DHCP server.
- **Source Node:** extClient1
- **Command:** ip a (or ifconfig)
- **Command:** dhclient -v eth0 (This command forces a new DHCP request. If dhclient is already running, it might just renew. Wait a few seconds for it to complete.)

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# dhclient -v eth0
Internet Systems Consortium DHCP Client 4.3.3
Copyright 2004-2015 Internet Systems Consortium.
All rights reserved.
For info, please visit https://www.isc.org/software/dhcp/

Listening on LPF/eth0/00:00:00:aa:00:1d
Sending on   LPF/eth0/00:00:00:aa:00:1d
Sending on   Socket/fallback
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 3 (xid=0x81ce983e)
DHCPDISCOVER on eth0 to 255.255.255.255 port 67 interval 6 (xid=0x81ce983e)
DHCPREQUEST of 114.181.172.102 on eth0 to 255.255.255.255 port 67 (xid=0x3e98ce81)
DHCPOFFER of 114.181.172.102 from 114.181.172.1
DHCPACK of 114.181.172.102 from 114.181.172.1
bound to 114.181.172.102 -- renewal in 1424 seconds.
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- Execute the command again: ip a (or ifconfig)

```

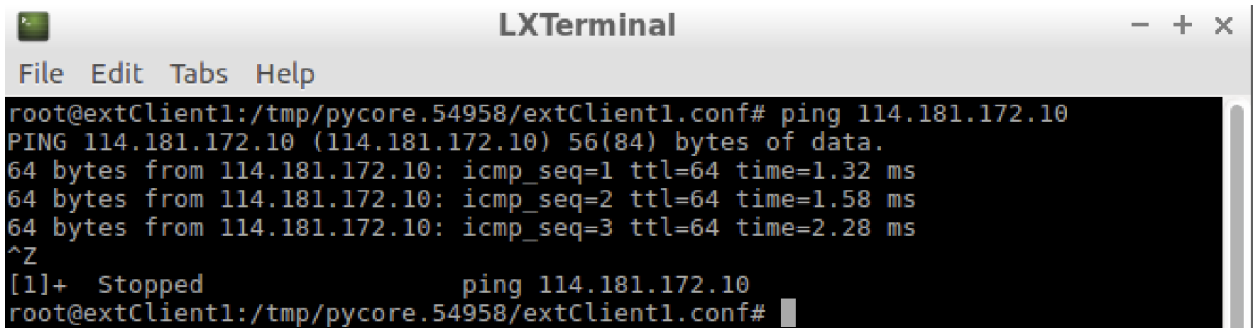
root@extClient1:/tmp/pycore[54958/extClient1.conf# ip a show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
1109: eth0@if1110: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether 00:00:00:aa:00:1d brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 114.181.172.20/24 scope global eth0
        valid_lft forever preferred_lft forever
    inet 114.181.172.100/24 brd 114.181.172.255 scope global secondary eth0
        valid_lft forever preferred_lft forever
    inet 114.181.172.102/24 brd 114.181.172.255 scope global secondary eth0
        valid_lft forever preferred_lft forever
    inet6 fe80::200:ff:feaa:1d/64 scope link
        valid_lft forever preferred_lft forever
root@extClient1:/tmp/pycore.54958/extClient1.conf#

```

- **Expected Results:**
- The eth0 interface on extClient1 (and extClient2) should have an IP address within the configured DHCP pool range (e.g., 114.181.172.100 to 114.181.172.200).
- The netmask should be 255.255.255.0 (/24).
- A broadcast address (e.g., 114.181.172.255) should be present.
- The dhclient command should show messages indicating a successful DHCPDISCOVER, DHCPOFFER, DHCPREQUEST, and DHCPACK process.

5. Verify Intra-Delos Network Connectivity (using DHCP and Static IPs)

- **Objective:** Confirm that all nodes within the Delos network can communicate, demonstrating successful IP assignment (dynamic and static) and local routing.
- **Source Node:** extClient1
- **Command:** ping 114.181.172.10 (Ieto) - Ping static server in same subnet.



```

LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# ping 114.181.172.10
PING 114.181.172.10 (114.181.172.10) 56(84) bytes of data:
64 bytes from 114.181.172.10: icmp_seq=1 ttl=64 time=1.32 ms
64 bytes from 114.181.172.10: icmp_seq=2 ttl=64 time=1.58 ms
64 bytes from 114.181.172.10: icmp_seq=3 ttl=64 time=2.28 ms
^Z
[1]+  Stopped                  ping 114.181.172.10
root@extClient1:/tmp/pycore.54958/extClient1.conf#

```

- **Command:** ping 114.181.172.1 (minerva) - Ping default gateway/router interface.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# ping 114.181.172.1
PING 114.181.172.1 (114.181.172.1) 56(84) bytes of data.
64 bytes from 114.181.172.1: icmp_seq=1 ttl=64 time=0.717 ms
64 bytes from 114.181.172.1: icmp_seq=2 ttl=64 time=1.61 ms
64 bytes from 114.181.172.1: icmp_seq=3 ttl=64 time=1.85 ms
^Z
[3]+  Stopped                  ping 114.181.172.1
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- **Command:** ping 114.181.127.11 (apollo) - Ping public server in other Delos subnet.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# ping 114.181.127.11
PING 114.181.127.11 (114.181.127.11) 56(84) bytes of data.
64 bytes from 114.181.127.11: icmp_seq=1 ttl=63 time=1.12 ms
64 bytes from 114.181.127.11: icmp_seq=2 ttl=63 time=1.70 ms
64 bytes from 114.181.127.11: icmp_seq=3 ttl=63 time=1.77 ms
^Z
[4]+  Stopped                  ping 114.181.127.11
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- **Command:** ping 114.181.127.10 (artemis) - Ping DNS server by IP.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# ping 114.181.127.10
PING 114.181.127.10 (114.181.127.10) 56(84) bytes of data.
64 bytes from 114.181.127.10: icmp_seq=1 ttl=63 time=0.600 ms
64 bytes from 114.181.127.10: icmp_seq=2 ttl=63 time=2.53 ms
64 bytes from 114.181.127.10: icmp_seq=3 ttl=63 time=0.775 ms
^Z
[5]+  Stopped                  ping 114.181.127.10
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- **Command:** ping 114.181.127.12 (demeter.delos.edu) - Ping public server by name (tests DNS resolution via artemis).

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# ping 114.181.127.12
PING 114.181.127.12 (114.181.127.12) 56(84) bytes of data.
64 bytes from 114.181.127.12: icmp_seq=1 ttl=63 time=1.39 ms
64 bytes from 114.181.127.12: icmp_seq=2 ttl=63 time=1.02 ms
64 bytes from 114.181.127.12: icmp_seq=3 ttl=63 time=2.02 ms
^Z
[6]+ Stopped ping 114.181.127.12
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- **Command:** traceroute 114.181.127.11 (apollo) - Check path to public server.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# traceroute 114.181.127.11
traceroute to 114.181.127.11 (114.181.127.11), 64 hops max
 1  114.181.172.1  0.109ms  0.076ms  0.047ms
 2  114.181.127.11 0.046ms  0.049ms  0.057ms
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- **Expected Results:**
- All ping commands should succeed.
- traceroute apollo should show hops through minerva's 114.181.172.1 interface, then to minerva's 114.181.127.x interface, and finally to apollo.

6. Verify Inter-Organization Connectivity

- **Objective:** Confirm that Delos clients (DHCP-assigned) can reach hosts in the Talos network and beyond, proving the DHCP-assigned default gateway and DNS work with the global routing.
- **Source Node:** extClient1
- **Command:** ping 181.56.181.11 (www.talos.edu) - Ping a public Talos server.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# ping 181.56.181.11
PING 181.56.181.11 (181.56.181.11) 56(84) bytes of data.
64 bytes from 181.56.181.11: icmp_seq=1 ttl=61 time=1.10 ms
64 bytes from 181.56.181.11: icmp_seq=2 ttl=61 time=1.60 ms
64 bytes from 181.56.181.11: icmp_seq=3 ttl=61 time=2.88 ms
^Z
[8]+  Stopped                  ping 181.56.181.11
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- **Command:** ping 181.56.18.10 (client1.talos.edu) - Ping a Talos internal host.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# ping 181.56.18.10
PING 181.56.18.10 (181.56.18.10) 56(84) bytes of data.
64 bytes from 181.56.18.10: icmp_seq=1 ttl=60 time=6.48 ms
64 bytes from 181.56.18.10: icmp_seq=2 ttl=60 time=4.08 ms
64 bytes from 181.56.18.10: icmp_seq=3 ttl=60 time=5.72 ms
^Z
[9]+  Stopped                  ping 181.56.18.10
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- **Command:** ping 196.37.86.10 (clio) - Ping the global DNS server.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# ping 196.37.86.10
PING 196.37.86.10 (196.37.86.10) 56(84) bytes of data.
64 bytes from 196.37.86.10: icmp_seq=1 ttl=62 time=1.54 ms
64 bytes from 196.37.86.10: icmp_seq=2 ttl=62 time=2.16 ms
64 bytes from 196.37.86.10: icmp_seq=3 ttl=62 time=1.71 ms
^Z
[10]+ Stopped                  ping 196.37.86.10
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- **Command:** traceroute 181.56.181.11 (www.talos.edu) - Observe the path to a Talos host.

```
LXTerminal
File Edit Tabs Help
root@extClient1:/tmp/pycore.54958/extClient1.conf# traceroute 181.56.181.11
traceroute to 181.56.181.11 (181.56.181.11), 64 hops max
 1  114.181.172.1  0.095ms  0.067ms  0.054ms
 2  48.198.245.1  0.053ms  0.055ms  0.053ms
 3  98.141.71.1  0.052ms  0.062ms  0.056ms
 4  181.56.181.11  0.055ms  0.057ms  0.060ms
root@extClient1:/tmp/pycore.54958/extClient1.conf#
```

- This traceroute output is correct and confirms that the traffic from extClient1 is being routed through minerva, then through the Internet network, and finally to web.talos.edu within the Talos network.

TASK C – Firewall

firewall.sh:

```
#!/bin/bash

# Flush all existing rules and delete user-defined chains
iptables -F
iptables -X
iptables -t nat -F
iptables -t nat -X
iptables -Z

# Set default policies to DROP (Requirement 8)
iptables -P INPUT DROP
iptables -P FORWARD DROP
iptables -P OUTPUT ACCEPT

#####
# Global: Allow established/related connections (stateful)
#####
iptables -A INPUT -m state --state RELATED,ESTABLISHED -j ACCEPT
iptables -A FORWARD -m state --state RELATED,ESTABLISHED -j ACCEPT
iptables -A OUTPUT -m state --state RELATED,ESTABLISHED -j ACCEPT

#####
# Rule 1: Public → DMZ for specific services only
# DNS: 181.56.181.10 (UDP/TCP 53)
# Web: 181.56.181.11 (TCP 80)
# Mail: 181.56.181.12 (TCP 25)
#####
iptables -A FORWARD -p udp -d 181.56.181.10 --dport 53 -i eth3 -o eth0 -j ACCEPT
```

```
iptables -A FORWARD -p tcp -d 181.56.181.10 --dport 53 -i eth3 -o eth0 -j ACCEPT
```

```
iptables -A FORWARD -p tcp -d 181.56.181.11 --dport 80 -i eth3 -o eth0 -j ACCEPT
```

```
iptables -A FORWARD -p tcp -d 181.56.181.12 --dport 25 -i eth3 -o eth0 -j ACCEPT
```

```
#####
```

```
# Rule 2: DMZ → External (only required services)
```

```
#####
```

```
iptables -A FORWARD -i eth0 -o eth3 -s 181.56.181.10 -p udp --dport 53 -j ACCEPT
```

```
iptables -A FORWARD -i eth0 -o eth3 -s 181.56.181.12 -p tcp --dport 25 -j ACCEPT
```

```
# Allow ICMP (ping) from DMZ to Internet for testing
```

```
iptables -A FORWARD -i eth0 -o eth3 -s 181.56.181.10 -p icmp --icmp-type echo-request -j ACCEPT
```

```
iptables -A FORWARD -i eth0 -o eth3 -s 181.56.181.11 -p icmp --icmp-type echo-request -j ACCEPT
```

```
iptables -A FORWARD -i eth0 -o eth3 -s 181.56.181.12 -p icmp --icmp-type echo-request -j ACCEPT
```

```
#####
```

```
# Rule 3: Internal → DMZ (allow all)
```

```
#####
```

```
iptables -A FORWARD -i eth1 -o eth0 -s 181.56.18.0/24 -d 181.56.181.0/24 -j ACCEPT
```

```
iptables -A FORWARD -i eth1 -o eth0 -s 181.56.40.0/24 -d 181.56.181.0/24 -j ACCEPT
```

```
iptables -A FORWARD -i eth2 -o eth0 -s 181.56.26.0/24 -d 181.56.181.0/24 -j ACCEPT
```

```
#####
```

```
# Rule 4: Internal ↔ Internal (only if traffic goes through R3)
```

```
#####
```

```
iptables -A FORWARD -i eth1 -o eth2 -s 181.56.18.0/24 -d 181.56.26.0/24 -j ACCEPT
```

```
iptables -A FORWARD -i eth1 -o eth2 -s 181.56.40.0/24 -d 181.56.26.0/24 -j ACCEPT
```

```
iptables -A FORWARD -i eth2 -o eth1 -s 181.56.26.0/24 -d 181.56.18.0/24 -j ACCEPT
```

```
iptables -A FORWARD -i eth2 -o eth1 -s 181.56.26.0/24 -d 181.56.40.0/24 -j ACCEPT
```

```
#####
```

```
# Rule 5: Internal → Internet (stateful)
```

```
#####
```

```
iptables -A FORWARD -i eth1 -o eth3 -s 181.56.18.0/24 -j ACCEPT
```

```
iptables -A FORWARD -i eth1 -o eth3 -s 181.56.40.0/24 -j ACCEPT
```

```
iptables -A FORWARD -i eth2 -o eth3 -s 181.56.26.0/24 -j ACCEPT
```

```
#####
```

```
# Rule 6: SSH to R3 from client subnet only (181.56.18.0/24)
```

```
#####
```

```
iptables -A INPUT -p tcp --dport 22 -s 181.56.18.0/24 -i eth1 -j ACCEPT
```

```
#####
```

```
# Rule 7: ICMP (ping) for R3, DMZ, and Internal
```

```
#####
```

```
# R3 → others (echo-requests)
```

```
iptables -A OUTPUT -p icmp --icmp-type echo-request -d 181.56.18.0/24 -j ACCEPT
```

```
iptables -A OUTPUT -p icmp --icmp-type echo-request -d 181.56.40.0/24 -j ACCEPT
```

```
iptables -A OUTPUT -p icmp --icmp-type echo-request -d 181.56.26.0/24 -j ACCEPT
```

```
iptables -A OUTPUT -p icmp --icmp-type echo-request -d 181.56.181.0/24 -j ACCEPT
```

```
# Internal/DMZ → R3 (echo-requests)
```

```
iptables -A INPUT -p icmp --icmp-type echo-request -s 181.56.18.0/24 -i eth1 -j  
ACCEPT
```

```
iptables -A INPUT -p icmp --icmp-type echo-request -s 181.56.40.0/24 -i eth1 -j  
ACCEPT
```

```
iptables -A INPUT -p icmp --icmp-type echo-request -s 181.56.26.0/24 -i eth2 -j  
ACCEPT
```

```
iptables -A INPUT -p icmp --icmp-type echo-request -s 181.56.181.0/24 -i eth0 -j  
ACCEPT
```

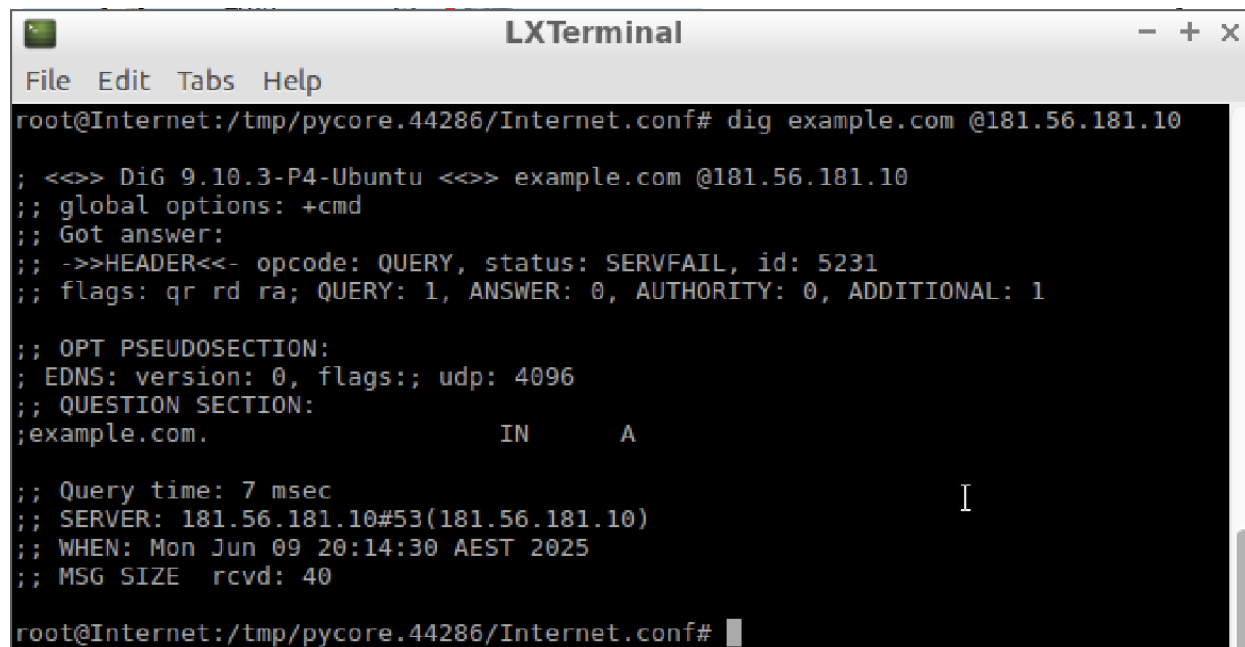
Internal ↔ DMZ via FORWARD (ping requests)

iptables -A FORWARD -p icmp --icmp-type echo-request -i e

TASK 3 - Testing

Requirement 1: Allow public (eth3) to DMZ (eth0) for specific services only

- **Test 1.1: Public to DNS (181.56.181.10) for UDP/53 (DNS query)**
 - **Source:** n8 (Internet)
 - **Command:** dig example.com @181.56.181.10
 - **Expected:** Success (DNS query resolves).



```
root@Internet:/tmp/pycore.44286/Internet.conf# dig example.com @181.56.181.10

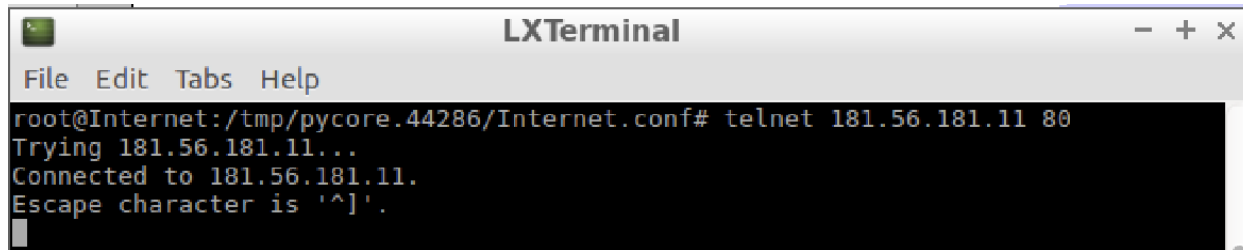
;<<>> DiG 9.10.3-P4-Ubuntu <<>> example.com @181.56.181.10
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 5231
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags;; udp: 4096
;; QUESTION SECTION:
;;example.com.                IN      A

;; Query time: 7 msec
;; SERVER: 181.56.181.10#53(181.56.181.10)
;; WHEN: Mon Jun 09 20:14:30 AEST 2025
;; MSG SIZE rcvd: 40

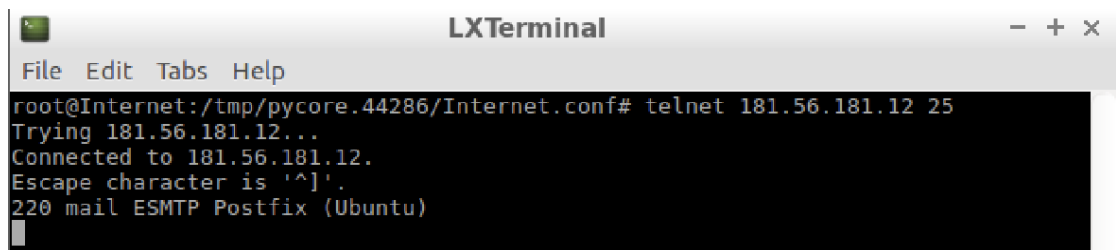
root@Internet:/tmp/pycore.44286/Internet.conf#
```

- **Test 1.2: Public to Web (181.56.181.11) for TCP/80 (HTTP)**
 - **Source:** n8 (Internet)
 - **Command:** curl http://181.56.181.11 or telnet 181.56.181.11 80
 - **Expected:** Success (HTTP response or telnet connection established).



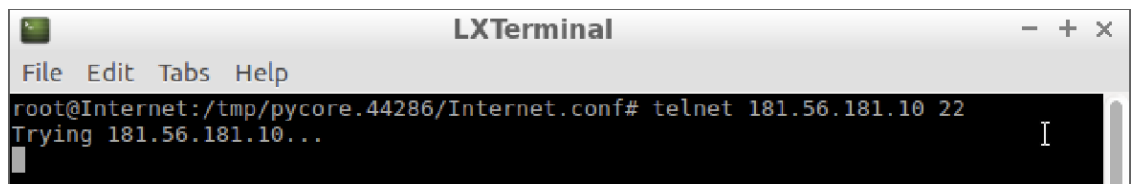
```
LXTerminal
File Edit Tabs Help
root@Internet:/tmp/pycore.44286/Internet.conf# telnet 181.56.181.11 80
Trying 181.56.181.11...
Connected to 181.56.181.11.
Escape character is '^['.
```

- **Test 1.3: Public to Mail (181.56.181.12) for TCP/25 (SMTP)**
 - **Source:** n8 (Internet)
 - **Command:** telnet 181.56.181.12 25
 - **Expected:** Success (Telnet connection established to SMTP banner).



```
LXTerminal
File Edit Tabs Help
root@Internet:/tmp/pycore.44286/Internet.conf# telnet 181.56.181.12 25
Trying 181.56.181.12...
Connected to 181.56.181.12.
Escape character is '^['.
220 mail ESMTP Postfix (Ubuntu)
```

- **Test 1.4 (Negative): Public to any other port on DMZ (e.g., SSH)**
 - **Source:** n8 (Internet)
 - **Command:** telnet 181.56.181.10 22 (assuming SSH is running but should be blocked)
 - **Expected:** Failure (Connection timed out).



```
LXTerminal
File Edit Tabs Help
root@Internet:/tmp/pycore.44286/Internet.conf# telnet 181.56.181.10 22
Trying 181.56.181.10...
```

Requirement 2:

Allow DMZ servers to initiate communication only if required by their service

- This includes **stateful access from DMZ** → **External**, limited to:
 - dns (181.56.181.10) → UDP 53 (can test with ping to Internet router)
- Command:** ping 98.141.71.2

```
LXTerminal
File Edit Tabs Help
root@dns:/tmp/pycore.44342/dns.conf# ping 98.141.71.2
PING 98.141.71.2 (98.141.71.2) 56(84) bytes of data.
64 bytes from 98.141.71.2: icmp_seq=1 ttl=63 time=0.975 ms
64 bytes from 98.141.71.2: icmp_seq=2 ttl=63 time=1.91 ms
^Z
[1]+  Stopped                  ping 98.141.71.2
root@dns:/tmp/pycore.44342/dns.conf#
```

- web (181.56.181.11) → *optional* (can test ping)

Command: ping 98.141.71.2

```
LXTerminal
File Edit Tabs Help
root@web:/tmp/pycore.44342/web.conf# ping 98.141.71.2
PING 98.141.71.2 (98.141.71.2) 56(84) bytes of data.
64 bytes from 98.141.71.2: icmp_seq=1 ttl=63 time=1.08 ms
64 bytes from 98.141.71.2: icmp_seq=2 ttl=63 time=0.884 ms
^Z
[1]+  Stopped                  ping 98.141.71.2
root@web:/tmp/pycore.44342/web.conf#
```

- mail (181.56.181.12) → TCP 25 (can test ping)

Command: ping 98.141.71.2

```
LXTerminal
File Edit Tabs Help
root@mail:/tmp/pycore.44342/mail.conf# ping 98.141.71.2
PING 98.141.71.2 (98.141.71.2) 56(84) bytes of data.
64 bytes from 98.141.71.2: icmp_seq=1 ttl=63 time=1.38 ms
64 bytes from 98.141.71.2: icmp_seq=2 ttl=63 time=0.500 ms
^Z
[1]+  Stopped                  ping 98.141.71.2
root@mail:/tmp/pycore.44342/mail.conf#
```

Requirement 3: Internal (eth1, eth2) → DMZ (eth0) (stateful, allow all)

- **Test 3.1: Internal Network 1 (n9 or n1) to DMZ (any service)**
 - **Source:** n9 (client1 - e.g., 181.56.18.150) or n1 (R1 - e.g., 181.56.40.1).
 - **Commands:**
 - `dig internal.example.com @181.56.181.10`

```
LXTerminal
File Edit Tabs Help
root@client1:/tmp/pycore.44286/client1.conf# dig internal.example.com @181.56.181.10
1.10

; <<> DiG 9.10.3-P4-Ubuntu <<> internal.example.com @181.56.181.10
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: SERVFAIL, id: 28682
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;internal.example.com.          IN      A

;; Query time: 8 msec
;; SERVER: 181.56.181.10#53(181.56.181.10)
;; WHEN: Mon Jun 09 20:19:25 AEST 2025
;; MSG SIZE rcvd: 49

root@client1:/tmp/pycore.44286/client1.conf#
```

- telnet 181.56.181.12 25

```
LXTerminal
File Edit Tabs Help
root@client1:/tmp/pycore.44286/client1.conf# telnet 181.56.181.12 25
Trying 181.56.181.12...
Connected to 181.56.181.12.
Escape character is '^]'.
220 mail ESMTP Postfix (Ubuntu)

```

- **Expected:** Success for all.
- **Test 3.2: Internal Network 2 (n2) to DMZ (any service)**
 - **Source:** n2 (R2 - 181.56.26.2).
 - **Commands:**
 - dig internal.example.com @181.56.181.10

```
LXTerminal
File Edit Tabs Help
root@R2:/tmp/pycore.44286/R2.conf# dig internal.example.com @181.56.181.10

; <<>> DiG 9.10.3-P4-Ubuntu <<>> internal.example.com @181.56.181.10
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: SERVFAIL, id: 23202
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 4096
;; QUESTION SECTION:
;internal.example.com.          IN      A

;; Query time: 4 msec
;; SERVER: 181.56.181.10#53(181.56.181.10)
;; WHEN: Mon Jun 09 20:42:10 AEST 2025
;; MSG SIZE rcvd: 49

root@R2:/tmp/pycore.44286/R2.conf#
```

- telnet 181.56.181.12 25

```
LXTerminal
File Edit Tabs Help
root@R2:/tmp/pycore.44286/R2.conf# telnet 181.56.181.12 25
Trying 181.56.181.12...
Connected to 181.56.181.12.
Escape character is '^]'.
220 mail ESMTP Postfix (Ubuntu)

```

- **Expected:** Success for all.

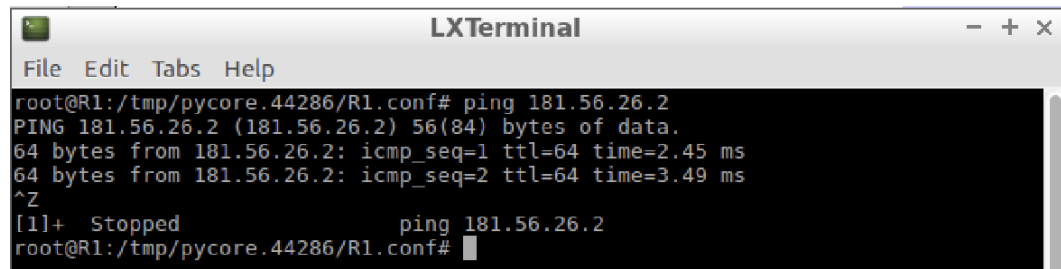
Requirement 4: Internal ↔ Internal (between networks on eth1 and eth2)

- **Test 4.1: Internal Network 1 (n9) to Internal Network 2 (n2)**
 - **Source:** n9 (client1 - e.g., 181.56.18.150).
 - **Command:** ping 181.56.26.2

```
LXTerminal
File Edit Tabs Help
root@client1:/tmp/pycore.44286/client1.conf# ping 181.56.26.2
PING 181.56.26.2 (181.56.26.2) 56(84) bytes of data.
64 bytes from 181.56.26.2: icmp_seq=1 ttl=63 time=6.18 ms
64 bytes from 181.56.26.2: icmp_seq=2 ttl=63 time=3.51 ms
^Z
[1]+  Stopped                  ping 181.56.26.2
root@client1:/tmp/pycore.44286/client1.conf#
```

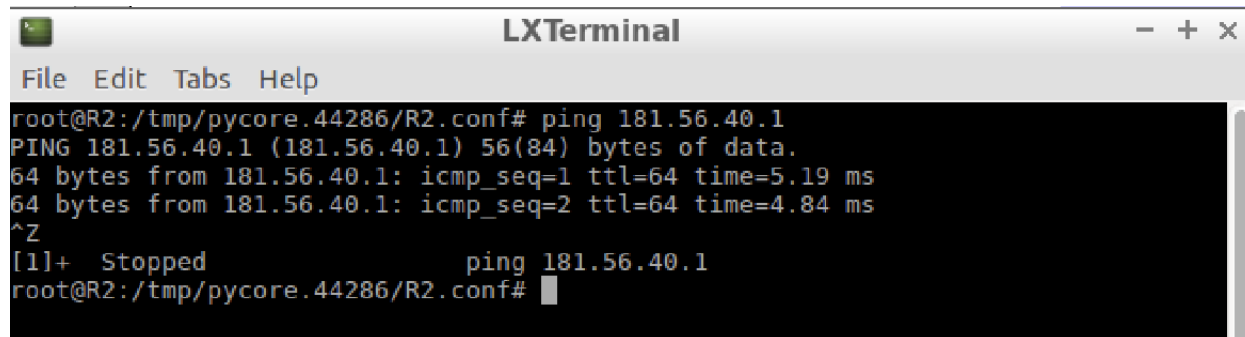
- **Expected:** Success.
- **Test 4.2: Internal Network 1 (n1) to Internal Network 2 (n2)**
 - **Source:** n1 (R1 - 181.56.40.1).

- **Command:** ping 181.56.26.2



```
LXTerminal
File Edit Tabs Help
root@R1:/tmp/pycore.44286/R1.conf# ping 181.56.26.2
PING 181.56.26.2 (181.56.26.2) 56(84) bytes of data.
64 bytes from 181.56.26.2: icmp_seq=1 ttl=64 time=2.45 ms
64 bytes from 181.56.26.2: icmp_seq=2 ttl=64 time=3.49 ms
^Z
[1]+  Stopped                  ping 181.56.26.2
root@R1:/tmp/pycore.44286/R1.conf#
```

- **Expected:** Success.
- **Test 4.3: Internal Network 2 (n2) to Internal Network 1 (n1)**
 - **Source:** n2 (R2 - 181.56.26.2).
 - **Command:** ping 181.56.40.1

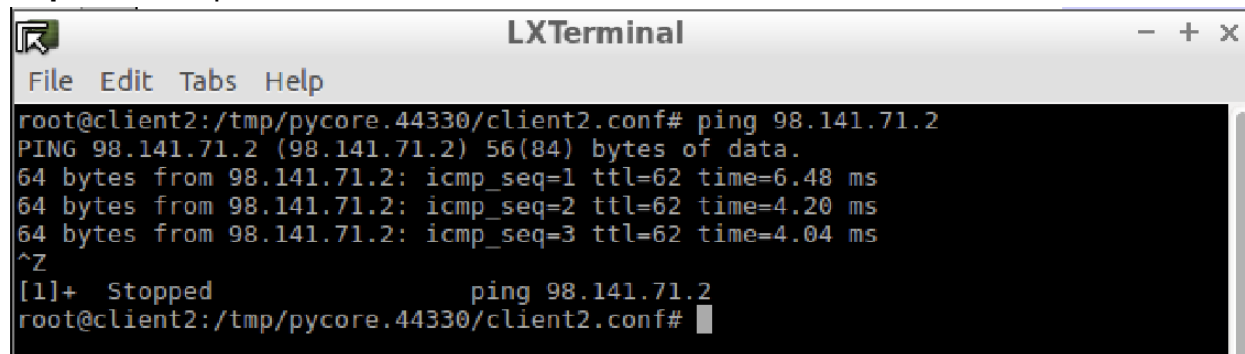


```
LXTerminal
File Edit Tabs Help
root@R2:/tmp/pycore.44286/R2.conf# ping 181.56.40.1
PING 181.56.40.1 (181.56.40.1) 56(84) bytes of data.
64 bytes from 181.56.40.1: icmp_seq=1 ttl=64 time=5.19 ms
64 bytes from 181.56.40.1: icmp_seq=2 ttl=64 time=4.84 ms
^Z
[1]+  Stopped                  ping 181.56.40.1
root@R2:/tmp/pycore.44286/R2.conf#
```

- **Expected:** Success.

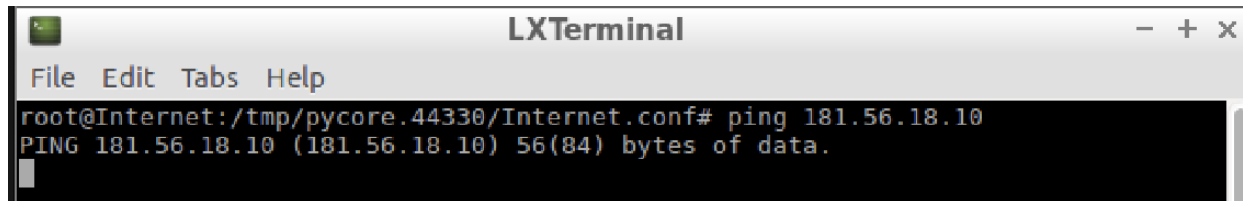
Requirement 5: Internal → Internet (stateful only)

- Internal can access Internet, but Internet cannot initiate to Internal
- **Test 5.1: Internal → Internet**
 - **Command:** ping 98.141.71.2
 - **Expected:** Replies from Internet router



```
LXTerminal
File Edit Tabs Help
root@client2:/tmp/pycore.44330/client2.conf# ping 98.141.71.2
PING 98.141.71.2 (98.141.71.2) 56(84) bytes of data.
64 bytes from 98.141.71.2: icmp_seq=1 ttl=62 time=6.48 ms
64 bytes from 98.141.71.2: icmp_seq=2 ttl=62 time=4.20 ms
64 bytes from 98.141.71.2: icmp_seq=3 ttl=62 time=4.04 ms
^Z
[1]+  Stopped                  ping 98.141.71.2
root@client2:/tmp/pycore.44330/client2.conf#
```

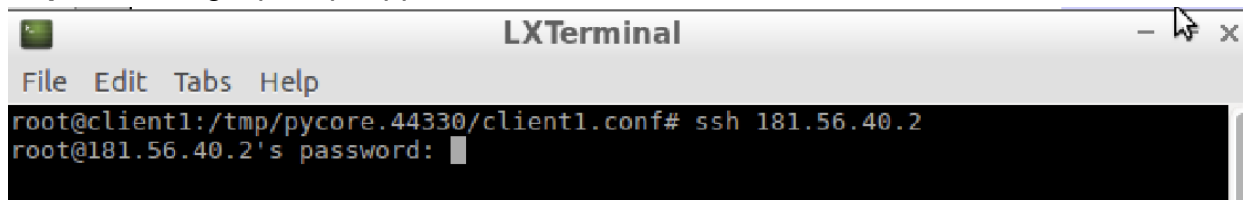
- **Test 5.2: Internet → Internal (blocked)**
 - **Command:** ping 181.56.18.10
 - **Expected:** No reply — incoming packets blocked



```
LXTerminal
File Edit Tabs Help
root@Internet:/tmp/pycore.44330/Internet.conf# ping 181.56.18.10
PING 181.56.18.10 (181.56.18.10) 56(84) bytes of data.
```

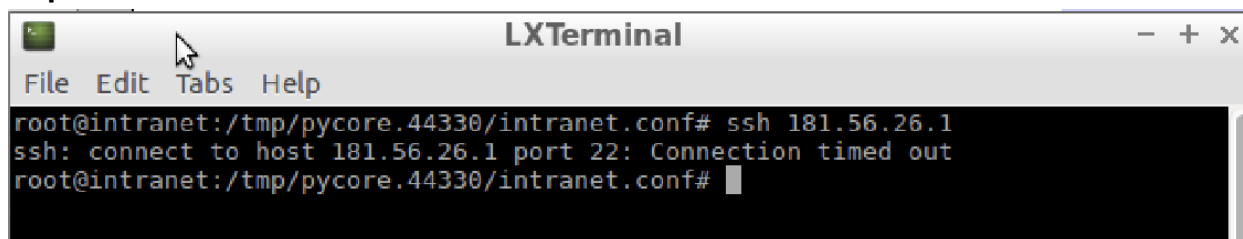
Requirement 6: SSH to R3 from clients only (181.56.18.0/24)

- **Test 6.1: SSH from client1 to R3**
 - **Command:** ssh 181.56.40.2
 - **Expected:** Login prompt appears



```
LXTerminal
File Edit Tabs Help
root@client1:/tmp/pycore.44330/client1.conf# ssh 181.56.40.2
root@181.56.40.2's password: 
```

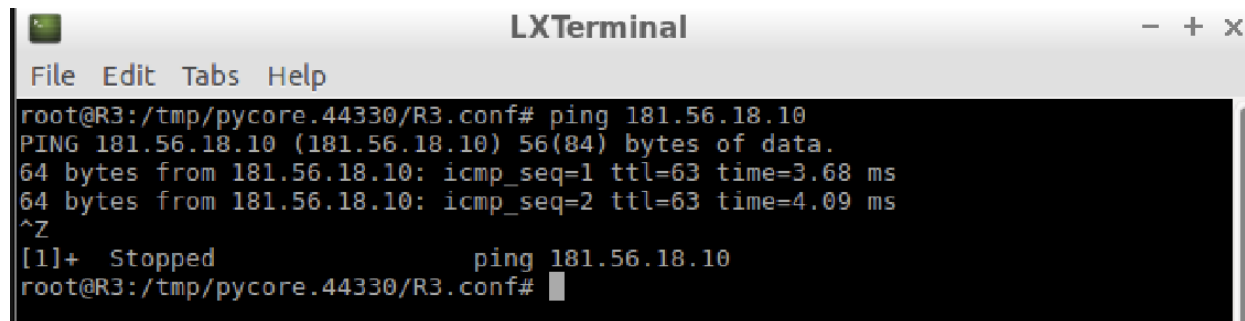
- **Test 6.2: SSH from intranet to R3**
 - **Command:** ssh 181.56.26.1
 - **Expected:** Connection refused/blocked



```
LXTerminal
File Edit Tabs Help
root@intranet:/tmp/pycore.44330/intranet.conf# ssh 181.56.26.1
ssh: connect to host 181.56.26.1 port 22: Connection timed out
root@intranet:/tmp/pycore.44330/intranet.conf# 
```

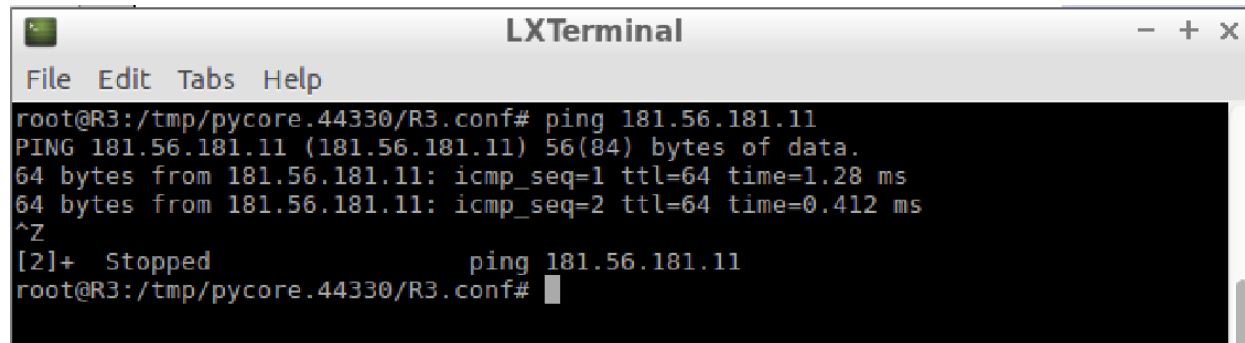
Requirement 7: ICMP (ping) between R3, Internal, DMZ

- **Test 7.1: R3 → client1**
 - **Command:** ping 181.56.18.10



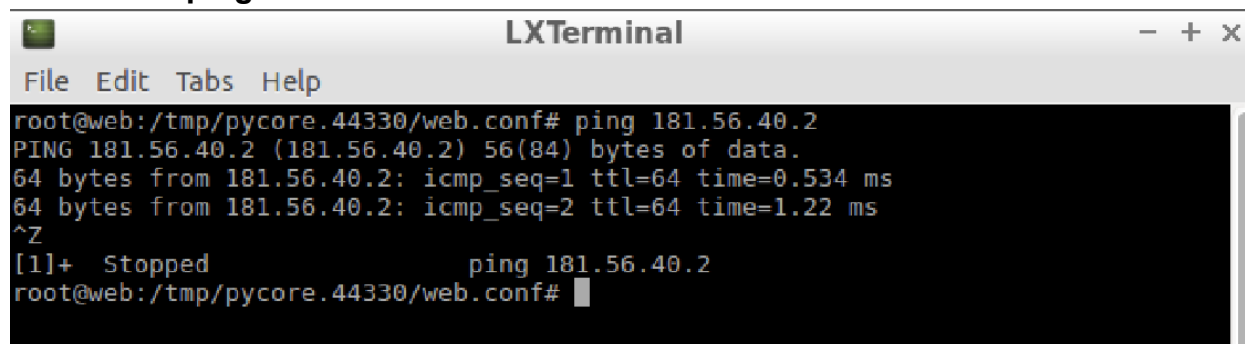
```
LXTerminal
File Edit Tabs Help
root@R3:/tmp/pycore.44330/R3.conf# ping 181.56.18.10
PING 181.56.18.10 (181.56.18.10) 56(84) bytes of data.
64 bytes from 181.56.18.10: icmp_seq=1 ttl=63 time=3.68 ms
64 bytes from 181.56.18.10: icmp_seq=2 ttl=63 time=4.09 ms
^Z
[1]+  Stopped                  ping 181.56.18.10
root@R3:/tmp/pycore.44330/R3.conf#
```

- **Test 7.2: R3 → web**
 - **Command:** ping 181.56.181.11



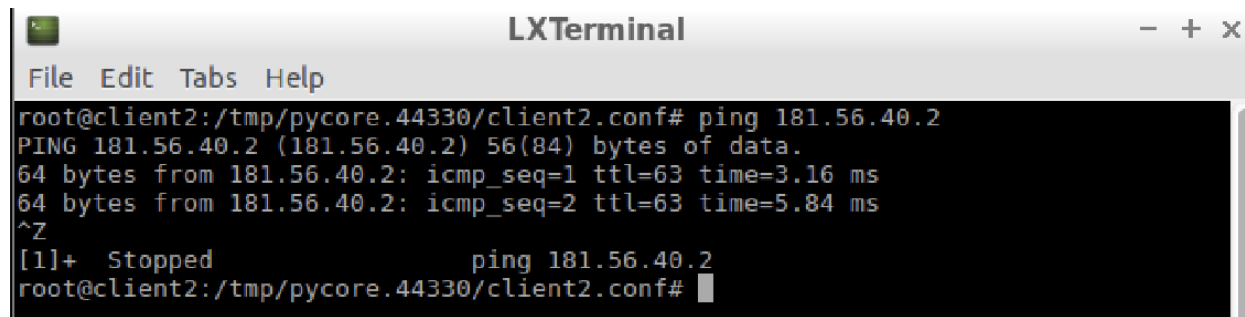
```
LXTerminal
File Edit Tabs Help
root@R3:/tmp/pycore.44330/R3.conf# ping 181.56.181.11
PING 181.56.181.11 (181.56.181.11) 56(84) bytes of data.
64 bytes from 181.56.181.11: icmp_seq=1 ttl=64 time=1.28 ms
64 bytes from 181.56.181.11: icmp_seq=2 ttl=64 time=0.412 ms
^Z
[2]+  Stopped                  ping 181.56.181.11
root@R3:/tmp/pycore.44330/R3.conf#
```

- **Test 7.3: web → R3**
 - **Command:** ping 181.56.40.2



```
LXTerminal
File Edit Tabs Help
root@web:/tmp/pycore.44330/web.conf# ping 181.56.40.2
PING 181.56.40.2 (181.56.40.2) 56(84) bytes of data.
64 bytes from 181.56.40.2: icmp_seq=1 ttl=64 time=0.534 ms
64 bytes from 181.56.40.2: icmp_seq=2 ttl=64 time=1.22 ms
^Z
[1]+  Stopped                  ping 181.56.40.2
root@web:/tmp/pycore.44330/web.conf#
```

- **Test 7.4: client2 → R3**
 - **Command:** ping 181.56.40.2

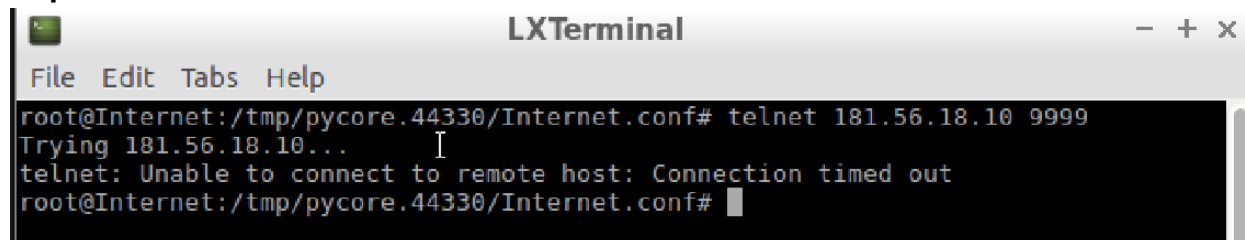
A screenshot of an LXTerminal window. The title bar says "LXTerminal" and has standard window controls. The menu bar shows "File Edit Tabs Help". The terminal text shows a user at a root prompt on client2 running a ping command to 181.56.40.2. The output shows two successful ping packets with 64 bytes of data, TTL of 63, and times of 3.16 ms and 5.84 ms. The user then presses Ctrl-Z (^Z) and the terminal shows "[1]+ Stopped ping 181.56.40.2" followed by the root prompt.

```
root@client2:/tmp/pycore.44330/client2.conf# ping 181.56.40.2
PING 181.56.40.2 (181.56.40.2) 56(84) bytes of data.
64 bytes from 181.56.40.2: icmp_seq=1 ttl=63 time=3.16 ms
64 bytes from 181.56.40.2: icmp_seq=2 ttl=63 time=5.84 ms
^Z
[1]+  Stopped                  ping 181.56.40.2
root@client2:/tmp/pycore.44330/client2.conf#
```

- **Expected:** All ICMP echo request + reply packets succeed

Rule 8: All Other Traffic Must Be Dropped

- **Test 8.1: Any random port from Internet to internal**
 - This requirement is implicitly verified by all the "Negative" tests above. Any successful connection or ping that was "Expected: Failure" would indicate a violation of this default drop policy.
 - **Command:** telnet 181.56.18.10 9999
 - **Expected:** Connection fails

A screenshot of an LXTerminal window. The title bar says "LXTerminal" and has standard window controls. The menu bar shows "File Edit Tabs Help". The terminal text shows a user at a root prompt on Internet running a telnet command to 181.56.18.10 port 9999. The output shows "Trying 181.56.18.10..." followed by "telnet: Unable to connect to remote host: Connection timed out". The user then presses Ctrl-C (indicated by a vertical bar) and the terminal shows the root prompt.

```
root@Internet:/tmp/pycore.44330/Internet.conf# telnet 181.56.18.10 9999
Trying 181.56.18.10...
telnet: Unable to connect to remote host: Connection timed out
root@Internet:/tmp/pycore.44330/Internet.conf#
```